



FORT STREET HIGH SCHOOL

Name: _____

Teacher: _____

Class: _____

2017 ASSESSMENT TASK 3 – PRELIMINARY

Mathematics Extension 1

Time allowed: 90 minutes

(plus 5 minutes reading time)

Syllabus Outcomes	Assessment Area Description and Marking Guidelines	Questions
P4	chooses and applies appropriate arithmetic, algebraic, graphical and trigonometric techniques	2,3,4
P6 P7	relates the derivative of a function to the slope of its graph determines the derivative of a function through routine application of the rules of differentiation	1, 5
PE6	makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations	5

General Instructions:

- Questions 1-5 are to be started in a new booklet
- The marks allocated for each question are indicated
- In Questions 1-5, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used

Total Marks 65

Attempt Questions 1-5

Question	Out of	Marks
Q1	/13	
Q2	/13	
Q3	/13	
Q4	/13	
Q5	/13	
Total	/65	
	Percent:	

Question 1 (13 marks) Use a SEPARATE writing booklet.

a) P (8, 8) is a point on the parabola $x^2 = 8y$. Find the equation of the tangent at P. **2**

b) Differentiate the following functions.

i) $f(x) = \frac{x^3 - 3x^2 + 1}{x^2}$ **1**

ii) $f(x) = \left(x + \frac{1}{x}\right)^2$ **2**

iii) $f(x) = \frac{3x^2 - 1}{4x - 2}$ **2**

iv) $f(x) = 2x^3 (3 + x^2)^2$ **2**

c) Find $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$ **1**

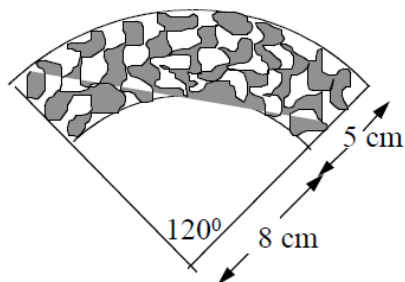
d) Find from first principles the derivative of $y = x^2 + 3x$ **3**

Question 2 (13 marks) Use a SEPARATE writing booklet.

a) Convert 210° into radians and leave your answer in terms of π

1

b) A fan in the shape of a sector of a circle is as follows



i) Find the perimeter of the shaded region, to the nearest cm?

2

ii) What area of material is used in the shaded part of the fan, to the nearest cm^2 ?

2

c) i) Show that $x = \frac{\pi}{3}$ is a solution of $\sin x = \frac{1}{2} \tan x$

1

ii) On the same set of axes, sketch the graphs of the function $y = \sin x$ and $y = \frac{1}{2} \tan x$

for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

2

iii) Using your graphs, or otherwise solve $\sin x \leq \frac{1}{2} \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

2

d) i) Show that $\operatorname{cosec} 2\theta - \cot 2\theta = \tan \theta$

2

ii) Hence find the exact value of $\tan \frac{\pi}{8}$

1

Question 3 (13 marks) Use a SEPARATE writing booklet.

- a) i) Express $\cos\theta + \sqrt{3} \sin\theta$ in the form $R \cos(\theta - \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ **2**
- ii) Sketch $y = R \cos(\theta - \alpha)$, $0 \leq \theta \leq 2\pi$. Showing all essential features. **2**
- iii) Solve the equation $\cos\theta + \sqrt{3} \sin\theta = \sqrt{2}$ for $0 \leq \theta \leq 2\pi$. **2**
- b) Find the general solution to $\cos 2\theta + \cos\theta + 1 = 0$, leaving your answer in exact form in radians. **3**
- c) Using the t results solve $1 + \cos x = 2 \sin^2 x$ for $0 \leq x \leq 2\pi$. **4**

Question 4 (13 marks) Use a SEPARATE writing booklet.

- a) Find the equation of the locus of a point P, which moves so it is always equidistant from the points A (3,5) and B (1,1). **3**

- b) Let A and B be the points (4,1) and (-2,2) respectively.

Find the equation of the locus of a point P, which moves so PA is perpendicular to PB.

Describe the locus geometrically. **3**

- c) For the parabola $y = \frac{1}{20} (x^2 - 8x - 4)$, find the **3**

i) Vertex

ii) Focus

- d) i) Find the equation of the parabola that has directrix $y = -4$ and cuts the x -axis at $x = -2$ and $x = 6$. **3**

ii) Hence, find the coordinates of the focus. **1**

Question 5 (13 marks) Use a SEPARATE writing booklet.

- a) Consider the function $f(x) = x^4 - 4x^3$.
- i) Show that $f'(x) = 4x^2(x - 3)$. **1**
 - ii) Find the coordinates of the stationary point(s) of the curve $y = f(x)$, and determine their nature. **3**
 - iii) Sketch the graph of the curve $y = f(x)$, showing the stationary point(s). **2**
 - iv) Find the values of x for which the graph of $y = f(x)$ is concave down. **2**
- b) i) Show that for all values of m , the line $y = mx - 3m^2$ touches the parabola $x^2 = 12y$. **2**
- ii) Find the values of m for which this line passes through the point (5, 2). **2**
- iii) Hence determine the equation of the tangent to the parabola $x^2 = 12y$, from the point (5, 2) when the value of m is an integer. **1**

End of paper

Question 1

Comments

$$a) \quad x^2 = 8y$$

$$y = \frac{x^2}{8}$$

$$f'(x) = \frac{x}{4}$$

$$f'(8) = 2 \quad \checkmark$$

The equation of the tangent at (8, 8)

$$y - y_1 = m(x - x_1)$$

$$y - 8 = 2(x - 8)$$

$$y = 2x - 8 \quad \checkmark$$

b)

$$i) \quad f(x) = \frac{x^3 - 3x^2 + 1}{x^2}$$

$$= x - 3 + x^{-2}$$

$$f'(x) = 1 - \frac{2}{x^3} \quad \checkmark$$

$$ii) \quad f(x) = \left(x + \frac{1}{x}\right)^2$$

$$f'(x) = 2\left(x + \frac{1}{x}\right)\left(1 - \frac{1}{x^2}\right) \quad \checkmark$$

$$iii) \quad f(x) = \frac{3x^2 - 1}{4x - 1}$$

$$f'(x) = \frac{v u' - u v'}{v^2}$$

$$\text{let } u = 3x^2 - 1$$

$$u' = 6x$$

$$v = 4x - 1$$

$$v' = 4$$

$$= \frac{(4x - 1)(6x) - (3x^2 - 1)(4)}{(4x - 1)^2} \quad \checkmark$$

$$= \frac{12x^2 - 6x + 4}{(4x - 1)^2} \quad \checkmark$$

$$iv \quad f(x) = 2x^3 (3+x^2)^2$$

$$\text{let } u = (3+x^2)^2$$

$$u' = 2(3+x^2)(2x)$$

$$v = 2x^3$$

$$v' = 6x^2$$

① correct derivatives of u and v

① correct answer using the product rule formula.

$$\begin{aligned} f'(x) &= u v' + v u' \\ &= (3+x^2)^2 (6x^2) + 2x^3 (4x)(3+x^2) \end{aligned}$$

$$= (3+x^2) [(3+x^2)(6x^2) + 8x^3]$$

$$= (3+x^2)(6x^4 + 8x^3 + 18x^2)$$

$$c \quad \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{(x-2)}$$

$$= 4 \quad \checkmark$$

$$d) \quad f(x) = x^2 + 3x$$

$$\begin{aligned} f(x+h) &= (x+h)^2 + 3(x+h) \\ &= x^2 + 2xh + h^2 + 3x + 3h \quad \checkmark \end{aligned}$$

$$\therefore f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + 3h + h^2}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{h(2x + 3 + h)}{h}$$

$$= 2x + 3 \quad \checkmark$$

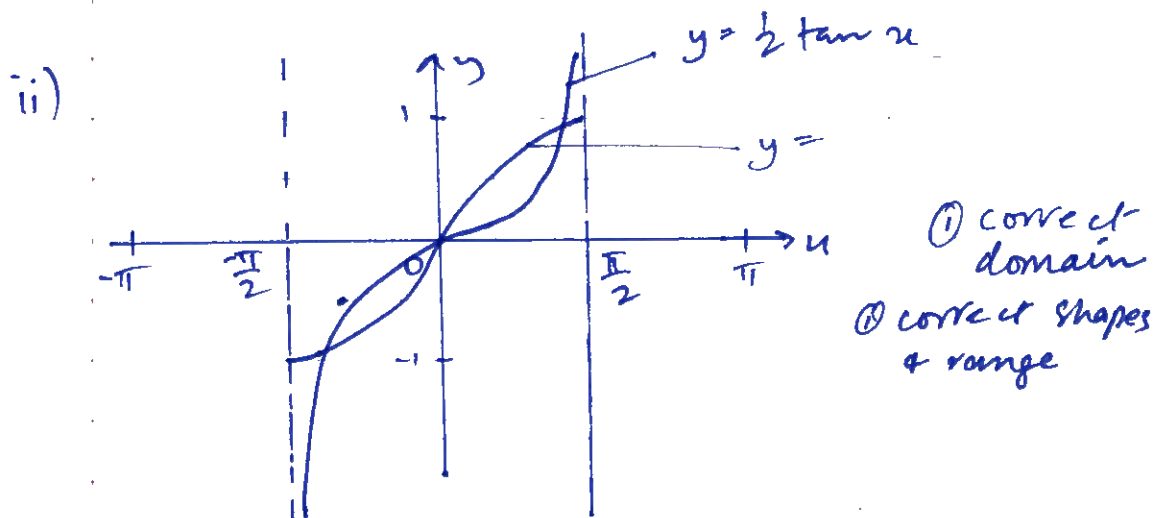
Question 2

a) $210^\circ \times \frac{\pi}{180} = \frac{7\pi}{6}$ ✓

b i) $P = 10 + r\theta + R\theta$ $(120^\circ \times \frac{\pi}{180} = \frac{2\pi}{3})$
 $= 10 + \left(\frac{2\pi}{3} \times 8 + \frac{2\pi}{3} \times 13\right)$
 $= 54 \text{ cm}$ ✓

ii) $A = \frac{1}{2} \theta (R^2 - r^2)$
 $= \frac{\pi}{3} (105)$ ✓
 $= 110 \text{ cm}^2$ ✓

c i) $\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$, $\frac{1}{2} \tan \frac{\pi}{3} = \frac{1}{2} \times \sqrt{3}$
 $= \frac{\sqrt{3}}{2}$ ✓
 $\therefore \text{LHS} = \text{RHS}$



iii) $-\frac{\pi}{6} \leq x \leq 0$ ✓

$\frac{\pi}{6} \leq x < \frac{\pi}{2}$ ✓

S	A
T	C

OR $\sin x \leq \frac{1}{2} \tan x$
 $\cos x \leq \frac{1}{2}$
 $x \leq \frac{\pi}{6}$

$\therefore -\frac{\pi}{6} \leq x \leq 0$

$$e) i) \operatorname{cosec} 2\theta - \cot 2\theta = \tan \theta$$

$$\text{LHS} = \frac{1}{\sin 2\theta} - \frac{\cos 2\theta}{\sin 2\theta}$$

$$= \frac{1 - \cos 2\theta}{\sin 2\theta}$$

$$= \frac{2\sin^2 \theta}{2\sin \theta \cos \theta}$$

$$= \frac{\sin \theta}{\cos \theta} \quad \checkmark \text{① ratio} \quad \text{① working}$$

$$= \tan \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$2\sin^2 \theta = 1 - \cos 2\theta$$

$$ii) \tan \frac{\pi}{8} = \operatorname{cosec} \frac{\pi}{4} - \cot \frac{\pi}{4}$$

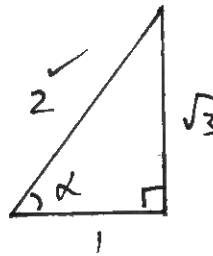
$$= \sqrt{2} - 1 \quad \checkmark$$

Question 3

a i) $\cos \theta + \sqrt{3} \sin \theta = R \cos(\theta - \alpha)$

$$\tan \alpha = \sqrt{3}$$

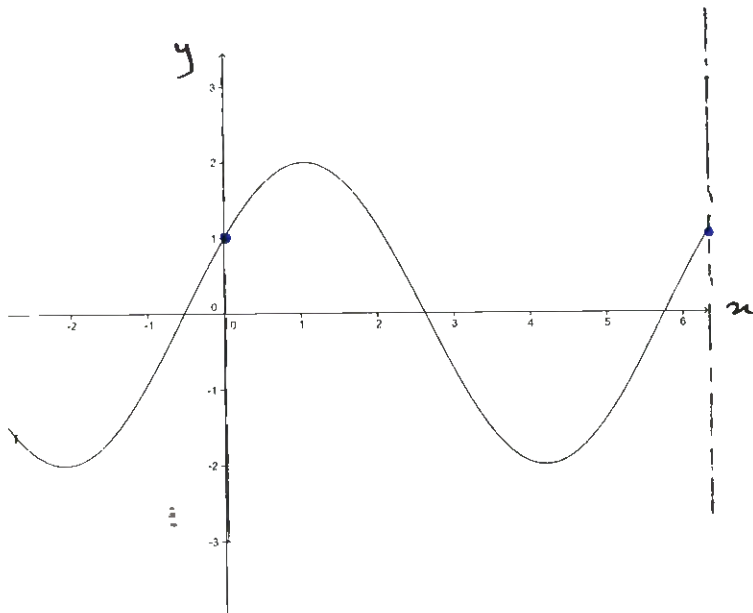
$$\alpha = \frac{\pi}{3} \checkmark$$



$$\therefore R \cos(\theta - \alpha) = 2 \cos\left(\theta - \frac{\pi}{3}\right)$$

ii) sketch:

$$0 \leq \theta \leq 2\pi$$



iii) $2 \cos\left(\theta - \frac{\pi}{3}\right) = \sqrt{2}$

$$\cos\left(\theta - \frac{\pi}{3}\right) = \frac{1}{\sqrt{2}}$$

$$\theta - \frac{\pi}{3} = \frac{\pi}{4} \checkmark, \quad \frac{7\pi}{4}$$

$$\theta = \frac{7\pi}{12}, \quad \frac{25\pi}{12} \checkmark$$

OR

$$\theta = \pm \frac{\pi}{4} + \frac{\pi}{3}$$

$$= \frac{7\pi}{12}, \quad \frac{\pi}{12}$$

(5)

3b

$$\cos 2\theta + \cos \theta + 1 = 0$$

$$2\cos^2 \theta - 1 + \cos \theta + 1 = 0$$

$$2\cos^2 \theta + \cos \theta = 0$$

$$\cos \theta (2\cos \theta + 1) = 0$$

$$\cos \theta = 0$$

$$\cos \theta = -\frac{1}{2}$$

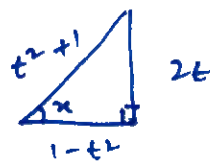
✓ for values

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$$

$$\theta = \frac{2\pi}{3}$$

$$\theta = 2\pi n \pm \frac{\pi}{2} \quad \checkmark$$

$$\theta = 2\pi n \pm \frac{2\pi}{3} \quad \checkmark$$

where n is an integer

$$c) \quad 2\sin^2 x - \cos x - 1 = 0$$

$$2\left(\frac{2t}{t^2+1}\right)^2 - \left(\frac{1-t^2}{t^2+1}\right) - 1 = 0 \quad \checkmark$$

$$8t^2 - [(1-t^2)(1+t^2)] - (t^2+1)^2 = 0$$

$$8t^2 - (1-t^4) - (t^4+2t^2+1) = 0$$

$$6t^2 - 2 = 0$$

$$t = \pm \frac{1}{\sqrt{3}} \quad \checkmark$$



$$\tan x = \frac{1}{\sqrt{3}}$$

$$\tan x = -\frac{1}{\sqrt{3}}$$

$$x = \frac{\pi}{6}, \frac{7\pi}{6} \quad \checkmark \quad x = \left(\pi - \frac{\pi}{6}\right), \left(2\pi - \frac{\pi}{6}\right)$$

$$= \frac{5\pi}{6}, \frac{11\pi}{6} \quad \checkmark$$

$$= -\frac{\pi}{6}, -\frac{7\pi}{6}$$

OR

(6)

Question 4

$$\begin{aligned}
 \text{a) } PA^2 &= PB^2 \\
 (x-3)^2 + (y-5)^2 &= (x-1)^2 + (y-1)^2 \\
 x^2 - 6x + 9 + y^2 - 10y + 25 &= x^2 - 2x + 1 + y^2 - 2y + 1
 \end{aligned}$$

$$\begin{aligned}
 -4x - 8y + 32 &= 0 \\
 x + 2y + 8 &= 0 \quad \checkmark
 \end{aligned}$$

$\therefore$ The locus of point P is a straight line

$$\begin{aligned}
 \text{b) } m_{PA} &= \frac{y-1}{x-4}, \quad m_{PB} = \frac{y-2}{x+2} \\
 \text{Since } PA &\perp PB && \text{for both gradients.} \\
 m_{PA} \times m_{PB} &= -1
 \end{aligned}$$

$$\frac{y-1}{x-4} = -\frac{(x+2)}{y-2} \quad \checkmark$$

$$(y-1)(y-2) = -(x+2)(x-4)$$

$$\begin{aligned}
 y^2 - 3y + 2 &= -(x^2 - 2x - 8) \\
 y^2 - 3y + 2 &= -x^2 + 2x + 8
 \end{aligned}$$

$$\begin{aligned}
 x^2 + 2x + y^2 - 3y - 6 &= 0 \quad \checkmark \\
 x^2 - 2x + (-1)^2 + y^2 - 3y &= 6 + 1 + \frac{9}{4}
 \end{aligned}$$

$$(x-1)^2 + \left(y - \frac{3}{2}\right)^2 = \left(\frac{\sqrt{37}}{2}\right)^2$$

The locus is a circle with centre $(1, \frac{3}{2})$ and radius is $\frac{\sqrt{37}}{2}$ $\checkmark$

$$c) \quad y = \frac{1}{20} (x^2 - 8x - 4)$$

$$x^2 - 8x - 4 = 20y$$

$$x^2 - 8x + (-4)^2 = 20y + 20$$

$$(x-4)^2 = 20(y+1) \quad \checkmark$$

$$4a = 20$$

$$a = 5$$

$$i) \quad \text{Vertex } (4, -1) \quad \checkmark$$

$$ii) \quad \text{Focus } (4, 4) \quad \checkmark$$

$$d) \quad (x-2)^2 = 4a(y-k)$$

$$\text{when } x=6, y=0$$

$$16 = -4ak \quad \text{--- (1)}$$

$$k-a = -4$$

$$k = a - 4 \quad \text{--- (2)}$$

Sub (2) into (1)

$$16 = -4a(a-4)$$

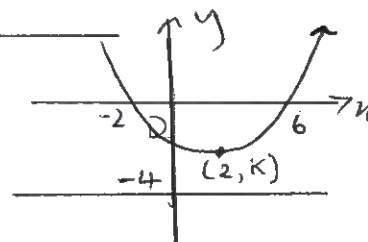
$$16 = -4a^2 + 16a$$

$$4a^2 - 16a + 16 = 0$$

$$a^2 - 4a + 4 = 0$$

$$(a-2)^2 = 0$$

$$a = 2 \quad \checkmark$$



from (2)

$$k = 2 - 4$$

$$= -2 \quad \checkmark$$

i) The equation of the parabola is:

$$(x-2)^2 = 8(y+2) \quad \checkmark$$

$$ii) \quad \text{Vertex: } (2, -2)$$

$$\text{Focus: } (2, 0) \quad \checkmark$$

Question 5

a) $f(x) = x^4 - 4x^3$

i) $f'(x) = 4x^3 - 12x^2$ ✓
 $= 4x^2(x-3)$

ii) for stationary points: $\frac{dy}{dx} = 0$

$$4x^2(x-3) = 0$$

$$x = 0, 3$$
 ✓

to determine their nature:

$$f''(x) = 12x^2 - 24x$$

$$f''(0) = 0 \quad (\text{a possible p.o.i.})$$

$$f''(3) = 36$$

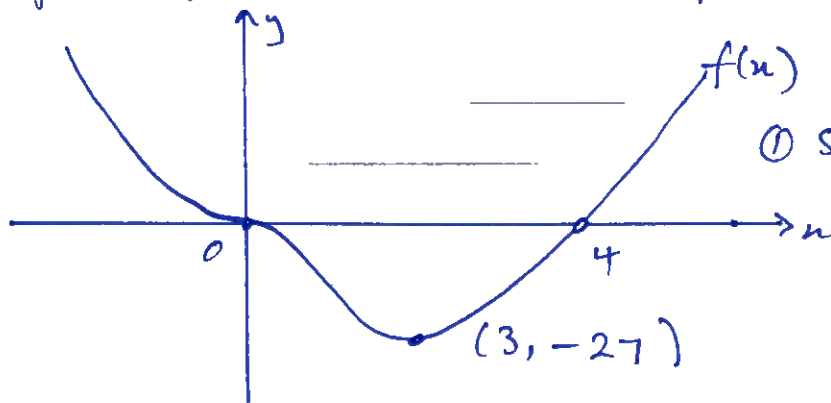
Since $f''(3) > 0$, $\therefore$ min when $x = 3$ ✓

x	-1	0	1	3	4
$f'(x)$	-16	0	-8	0	64
	-	0	-	0	+

* the table of values must include the $f'(x)$ values.
 + and - signs are not accepted in the HSC.

$\therefore$ from $f'(x)$, $x = 0, 3$ are p.o.i. ✓

iii)



- ① Shape + x intercepts
- ① minimum point.

iv) $f(x)$ is concave downward where $f''(x) < 0$

$$0 \leq x \leq 2$$
 ✓

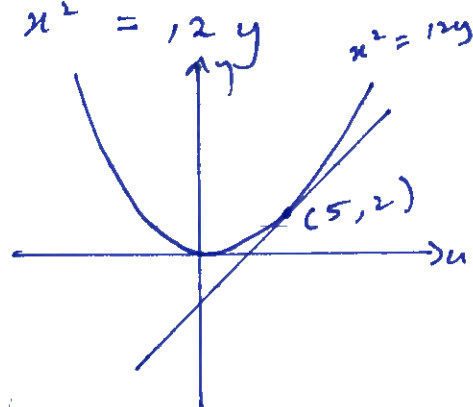
$$12x^2 - 24x < 0$$
 ✓



9

b) $y = mx - 3m^2$

, $x^2 = 12y$



i) $x^2 = 12(mx - 3m^2)$

$x^2 = 12mx - 36m^2$

for y to be a tangent

$\Delta = 0$

$b^2 - 4ac = 0$

$\Delta = (12m)^2 - 4(1)(36m^2)$

$= 144m^2 - 144m^2$

$= 0 \quad \checkmark$

ii) when passes through $(5, 2)$

$y = mx - 3m^2$

$2 = 5m - 3m^2 \quad \checkmark$

$3m^2 - 5m + 2 = 0$

$(3m+1)(m-2) = 0$

$\therefore m = 2, -\frac{1}{3} \quad \checkmark$

$\begin{matrix} 3m & 1 \\ m & -2 \end{matrix} \quad \times$

iii) The equation of the tangent is

$y = mx - 3m^2 \quad (\text{for } m = 2)$

$y = 2x - 12 \quad \checkmark$

END